# NETRIK HACKATHON 2026

## Loan Risk Prediction System – Documentation

## 1.Introduction

The loan prediction System is a web-based application developed to predict whether a loan application will be approved or not approved based on applicant details.

This system integrates:

- Frontend(HTML, CSS, JavaScript)
- Backend API(FastAPI)
- Machine learning model(Random Forest)
- Data Processing pipeline(Python)

Additionally, the system provides a reason-based response, explaining why a loan was approved or rejected.

## 2. Objective :

The main objectives of this project are :

- To automate the loan approved prediction process
- To build a user-friendly web interface
- To preprocess and clean real-world loan dotasets
- To train a machine learning classification model
- To integrate frontend and backend using REST API

- To provide both prediction and reason for decision

# 3. System Architecture :

## The system consists of two major parts:

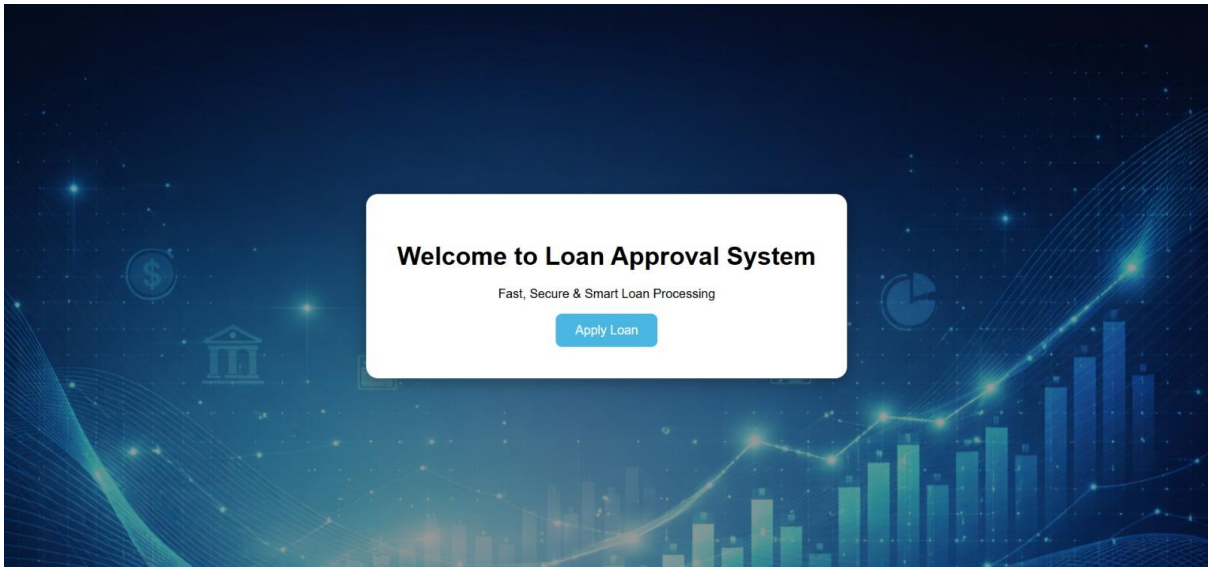
Frontend(User Interface)

**Developed using :**

- HTML
- CSS
- JavaScript

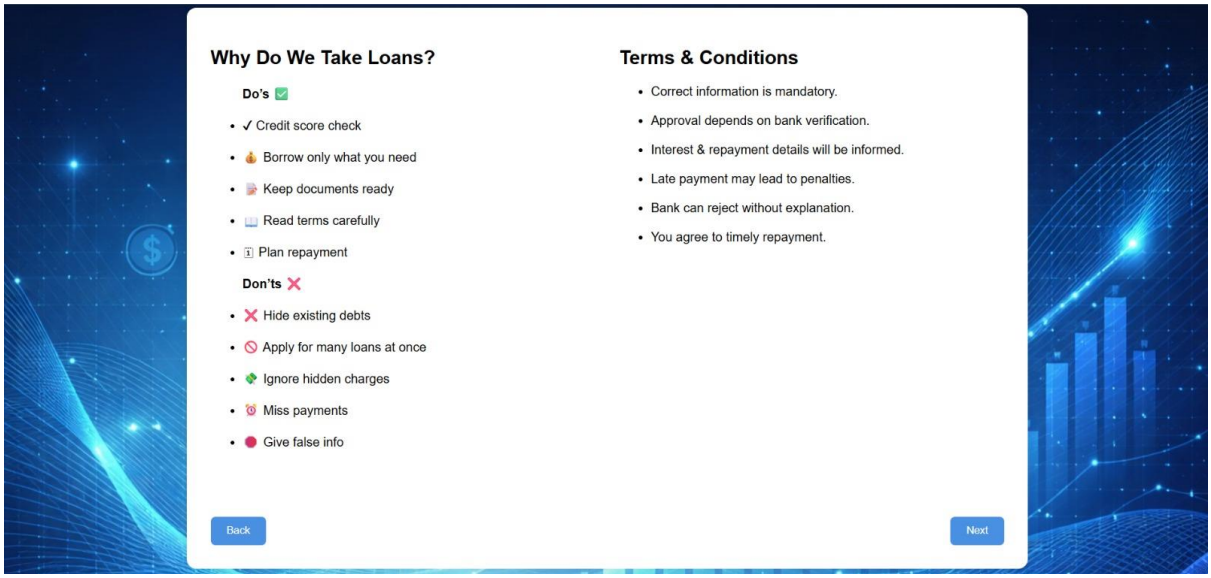
The frontend contains four connected pages:

1 1.home.html




- Displays welcome message
- Background image added
- “Apply Loan” button redirects to terms page

2.terms.html

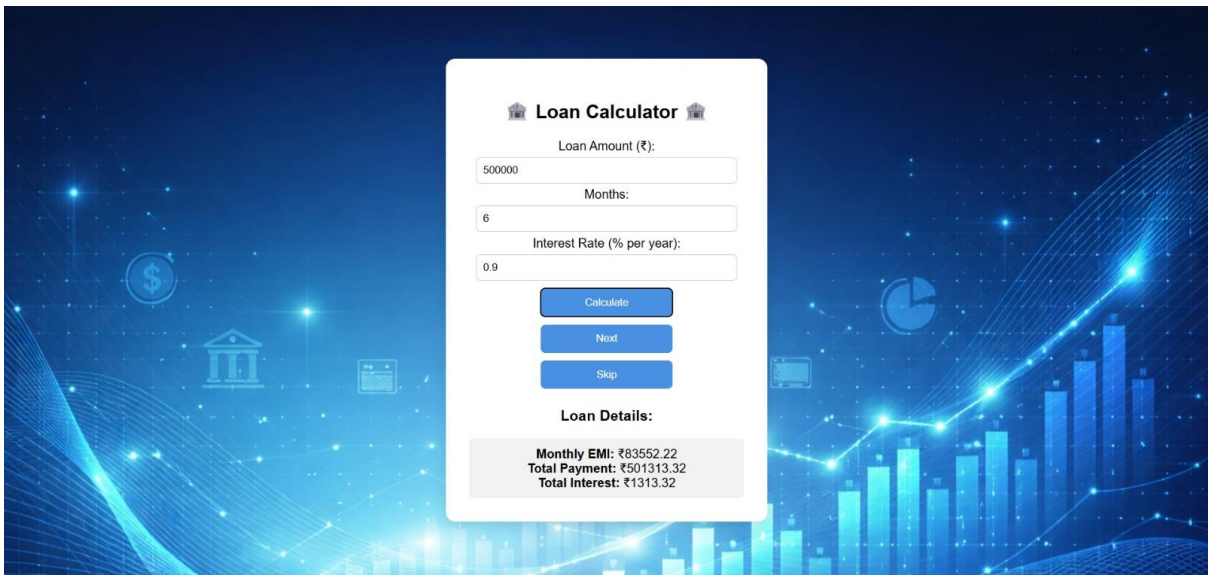



- Displays Do’s and Don’ts of Loan
- Displays Terms & conditions
- Navigation buttons:

  ->Back -> home.htlm

  ->Next -> cal.html

## 3.cal.html




- Calculators EMI

- Inputs :

  ->Loan Amount

  ->Number of months

  ->Interest Rate

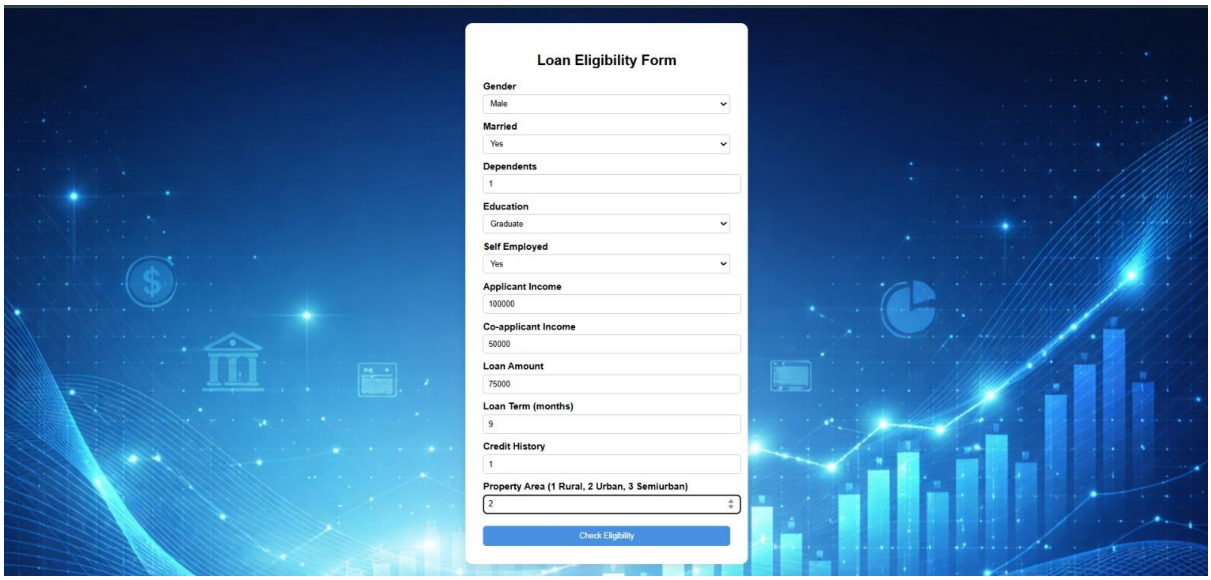
- Displays:

  ->Monthly EMI
  ->Total payment
  ->Total Interest

- User must calculate before proceeding

- Redirects to man.html

## 4.main.html(Loan Eligibility Form)




Collects applicant information:

- Gender

- Married

- Dependents

- Education
- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Term
- Credit History
- Properly Area

After submission:

- Data is sent to backend using:

**POST** :- http://127.0.0.1:8000/predict

Result is shown as alert message.

# 4.Data Processing Pipeline(python)

## Data Export (file.csv)

Dataset saved as CSV file using:
df.to_csv(...)

## Data Loading (data.py)

- Reads dataset using Pandas
- Displays first 5 rows

### Data Merging (Merge.py)

Three datasets are merged using Loan_ID :

- Dataset 1 -> Gender, ApplicationIncome
- Dataset 2 -> Credit_History, LoanAmount
- Dataset 3 -> Property_Area

Merged dataset saved as :
Merged.csv

### Data Cleaning(clean.py)

Cleaning steps performed:

- Missing LoanAmount filled using median
- Missing credit_History filled using mode
- Duplicate records removed
- Cleaned file saved as:

Cleaned.csv

## 5. Model Training(Train.py)

Algorithm used :
Random Forest Classifier

**Training Steps:**

- Load dataset
- Remove Loan_ID column
- Fill missing numeric values using median
- Fill missing categorial values using mode

- Convert Loan_Status:

  Y -> 1

  N ->0

- Apply one-Hot Encoding using get_dummies()

- Split dataset:

  80% Training

  20% Testing

- Train RandomForestClassifier

- Evaluate using:

  Accuracy

  Confusion Matrix

  Classification Report

- Save:

  Loan_model.pkl

  Loan_column.pkl

## 6.Backend API(Loan.py)

Developed using:

- FastAPI

- Pydantic

- Pandas

- Joblib
- Asyncio
- ThreadPoolExecutor

CORS enabled for frontend-backend communication.

## API is running

### GET "/"

Response:

Loan Prediction API Running

### POST "/predict"

Accepts applicant data in JSON format

Exampe Input:

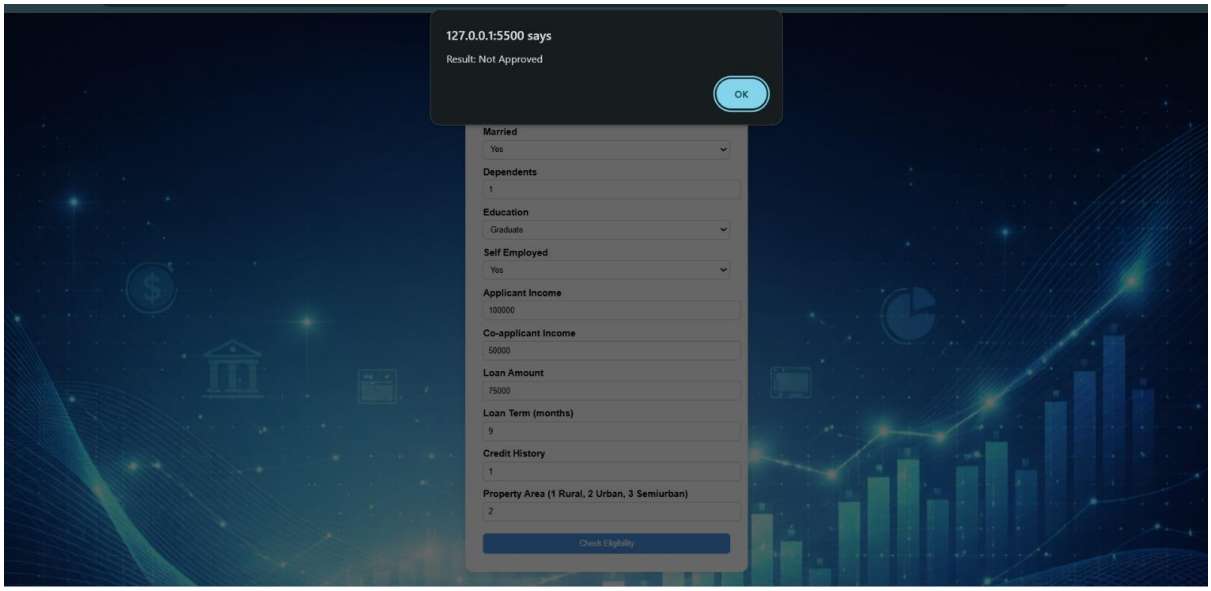
```
{
  "Gender": 1,
  "Married": 1,
 ..
}
```

## Prediction Logic

Steps:

- Convert input dictionary into DataFrame
- Apply one-hot encoding
- Reindex columns using saved column list
- Predict using trained model
- Generate result

## Reason-Based Output(Important Update)




The system not only predicts approval status but also gives a reason

<u>If Approved:</u>

- Returns:

Status: Approved

Reason: Strong financial profile and good credit history

<u>If Not Approved:</u>

- System checks conditions like:

  -> Poor credit history

  ->Low applicant income(< 2500)

  -> High loan amount (>250)

  ->Very long loan term(> 360 months)

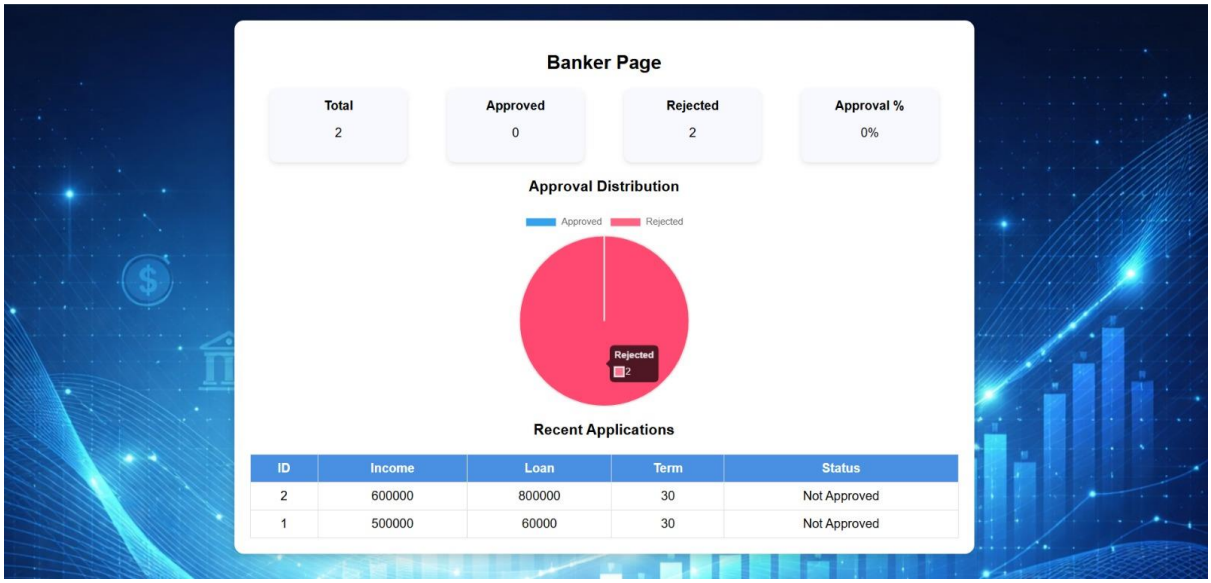
- Returns:

  ->Status: Not Approved

  ->Reason: List of detected risk factors

This makes the system more transparent and user-friendly.

# 7.Working Flow




This page is used by banker/admin to verify and monitor loan applications.

User ->
Home Page ->
Terms Page ->
Loan Calculator ->
Backend API ->
Machine Learning Model ->
Prediction + Reason ->
Result Displayed

# 8.Technologies Used

| **Component** | **Technology** |
|---|---|
| Frontend | HTML, CSS, JavaScript |
| Backend | FastAPI |
| ML Model | Random Forest |
| Data Processing | Pandas |
| Model Storage | Joblib |
| Async Handling | Asyncio + ThreadPoolExecutor |

# 9.Key Features

- Multi-page interactive frontend
- EMI calculator before eligibility check
- Data merging and cleaning pipeline

- Random Forest Classification model
- REST API integration
- Async prediction handling
- Reason-based loan display

# 10.Conclusion

The loan prediction system successfully integrates frontend development, backend API, and machine learning into a complete working application.

The system automates loan eligibility checking and provides both prediction and reason, making the decision process transparent and efficient.

This project demonstrates practical implementation of:

- Data preprocessing
- Machine learning model training
- REST API development
- Frontend-backend integration